

The mobile phone companies are racing against time to bring the changes about the mobile technology even faster. At present mobile phone 5G technology is a reality. It has been introduced in many countries of the world. In our country too it is going to be started.

2.5 Computer networking

Almost all of us are familiar with the term networking. In business, politics or sales you need proper communication with your clients and potential investors thus creating a network. Likewise, the communication and data transfer methods of two or more computing devices is jointly called computer networking. There are some sets of media and network devices needed to create a computer network.

2.5.1 Concept of Networking

We create networks among ourselves to make our daily lives easier, to socialize and to rely on other people when we alone can not make it. The evolution of information technology has enhanced and spread our networking abilities to a completely new dimension. Now we communicate through mobile phones and computers, via text messages, voice and video calls. We use wireless modes of connections while exchanging information between computer devices. So a computer network means a set of interconnected computer devices, bound by geographical boundaries and the process they use to communicate between each other. Using a computer network we can make bank transactions, send emails, receive government services and so on. Even if a computer in the network becomes unusable, another device in the network can provide support for it. A printer connected to a computer network can be used from every device on the network. Thus, the network makes the maximum use of every device present in the system.

2.5.2 Objectives of Computer Networking

The purpose of establishing a network between two or more devices is to share resources and communicate securely. A resource is an element or service in a network that is provided by some of the components of the network. Resource sharing between computers removes the limitations of data storage, data processing and makes every task faster and more effective. In short, resource sharing means -

Information Resource Sharing: The Internet is the best example of sharing information resources through a network. Locally connected computers can also share their internal information with LAN softwares.

Software Resource Sharing: Software resources can be shared via a computer network. Using this, only one software license can be bought and shared among the devices of the network so that every device of the network can use the software. In banking we see everyone using different computer terminals, but actually they are using the same software suits through networking.

Hardware Resource Sharing: We see offices, cyber cafes, banks and other organizations use network printers, where all the devices in the network can send print commands to one single printer. This is a really cost effective solution for any organization and makes the best use out of the device.

2.5.3 Types of networking

Boundless and unlimited supply of information is a basic need for the modern concept of globalization. To prioritize this concept of information exchange, the whole world has set up innumerable computer networks of different size and topography. These networks use a multitude of different network devices.

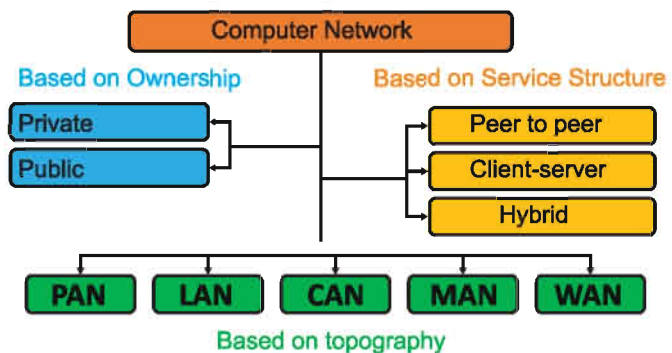


Figure 2.24: Types of Networks

Depending on the devices used or the topography, computer network can be classified according to the following criterions -

- Topography of the network
- Service structure and management of the network
- Ownership of the network

Topography of the Network

Depending on the geographical location of the devices inside a network, computer networks can be classified into five types -

1. Personal Area Network
2. Local Area network
3. Campus Area Network
4. Metropolitan Area Network
5. Wide Area Network

- 1. Personal Area Network:** This refers to the home network an individual person builds around his computer device, connecting printers, routers, phones, cameras and computers inside his/her home. The span of a PAN coverage can be 10 metres.



Figure 2.25: PAN

- 2. Local Area Network:** LAN is the most used network in our day to day lives. Offices, businesses, educational institutions use this type of networking to connect all the computers inside a building or one or more floors of the building. Using routers and repeaters the range of a LAN can be increased to a maximum of 1 kilometer. LAN can be of BUS, Star, Tree or Ring Topology. Twisted pair cables, coaxial cables, fiber optic cable or wireless media are used to create this type of network.



Figure 2.26: LAN

- 3. Campus Area Network:** CAN is made of multiple LANs in an area. Generally inside a wide educational institution area like a University uses CAN to connect their different buildings LAN systems. CAN can have a coverage range of between 1 to 5 kilometers. Large office complexes also use CANs to synchronize between devices inside their network of buildings.



Figure 2.27: CAN

- 4. Metropolitan Area Network:** A MAN connects the computer devices inside an urban or metropolitan area. MAN is bigger than a LAN but smaller than a WAN. MAN can have a coverage area of upto 50 kilometers. If the devices inside a MAN are connected via wireless media, it is called a WMAN or Wireless Metropolitan Area Network. Phone Lines, Optical Fiber Cable, Radio Waves or terrestrial microwaves are used as transmission media inside a MAN. Routers, switches, hubs, bridges and gateways are used as network devices.



Figure 2.28: MAN

- 5. Wide Area Network:** WAN connects different geographical areas networks into one. The internet is the prime example of a WAN.

Service and Management Structure

Depending on how the devices inside a network are managed and their service models, computer network can be classified into the following categories -

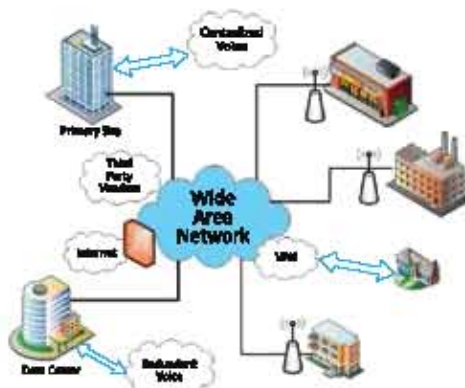


Figure 2.29: WAN

- 1. Peer to Peer Network:** Peer to peer network works without any central servers. Two or more computer devices are connected in the network and they can share information and resources between each other.
- 2. Client-Server Network:** This network contains two or more client stations and a centralised server station. The server computer stores the data of the network centrally and the data resources are shared between the client stations in the network. This is also called a server based network. Depending on the storage media, number of host and terminal nodes client-server networks can be of two types -

- a. Centralised Network: In this system only one centralised server is present is the server which is called the host. The host computer connects with multiple terminals.
- b. Distributed Network: This type of network connects multiple workstations or terminals, shared storage devices and I/O devices.

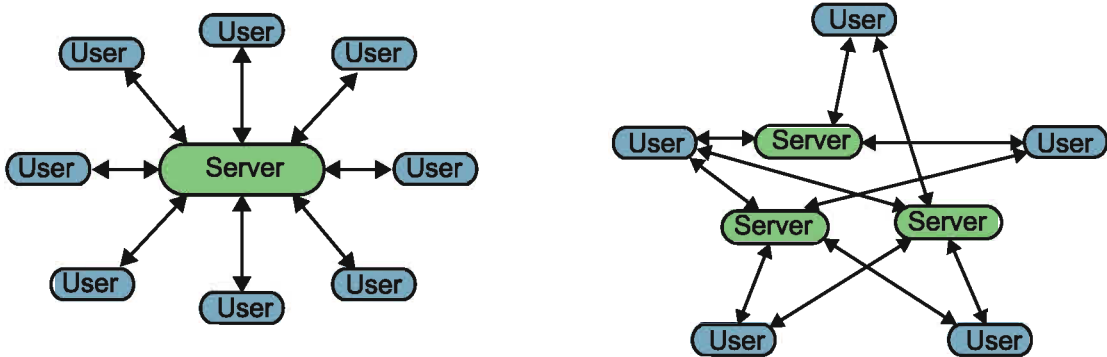


Figure 2.30: Centralised and Distributed Network

3. Hybrid Network: This type of network is a combination of peer to peer and client-server networks. In this system a host computer manages the network while it has the characteristics of a distributed network like shared storage media. This type of network is popular in corporate organizations. A hybrid network prioritises client servers.

Ownership of Network

Computer networks can be classified into two types depending on the ownership of the network and access and management systems.

1. Public Network: Public network does not have a restriction on network users so any computer device can join or leave the network at any time. This type of network usually is not owned by a particular individual or a company, rather it has the ownership of multiple parties. The users have to pay a certain fee for using this type of network. WAN or the internet is a good example of a public network.
2. Private Network: In private networks the number of users in the network is restricted and defined. Any new users wishing to join the network needs access permission from the network administrator. These types of networks are usually owned by an individual or a company. It has a better security system and less traffic than a public network. PAN < LAN or CANs are private networks.

2.5.4 Network Devices

The devices which are needed to construct and install a computer network are known as network devices. These devices work to transmit and receive the data, provide security and manage the flow and direction of the data flow. Some of the Important network devices are -

- **Modem:** Modems were invented before computer networks came about. They were used for creating telephone and television networks. The word Modem is the short form of Modulation and Demodulation. As we can guess from the name, this device converts analog signals to digital and vice versa. With the advent of optical fibers and wireless communication Modem has become almost obsolete.
- **Hub:** Hubs were used to connect a computer device to others in a network. If multiple computer devices are connected with wires in a hub, it will create a LAN. Hubs have no processing abilities. Hubs just broadcast a signal sent to the hub without any kind of alterations. For peer to peer communications all other devices except the destination devices refrain from receiving the signals. Hub does not have a collision protection system or data traffic management system. Hubs are also almost obsolete in the present world.
- **Switch:** Switches have replaced Hubs in computer networking systems. The switches perform the same tasks as a hub but it has processing powers and ability. Switch does not broadcast a signal it uses Media Access Control or MAC address of a computer to uniquely identify it inside the switch and transmit the data to that particular device port, thus avoiding collision. It can also amplify weak signals. Switches can have 8,16,24 or 48 ports. Switches can be used to filter data but the process is a bit complex. Switches are used to create LANs but it cannot create multiple LANs.
- **Router:** A router is a connecting device that can connect two or more networks with the same protocol and extend a network's coverage. Routers can connect multiple LANs to create a WAN. Routers use Network Address Translation or NAT to identify a network. Routers can determine the shortest path for a



Figure 2.31: Switch

transmission of data to one device of one network to another device in a different network. Routers can filter data. They can sense an increase of traffic in a network and use a different route to transmit data.

Configuring a router is a bit complicated. Routers can only connect networks which have the same protocols. It cannot connect networks with different protocols.



Figure 2.32: Router

- **Gateway:** Gateways are used to connect two networks with different protocols. This is a network connectivity device that connects two or more networks irrespective of their protocols. This device is a bit expensive and harder to configure. But gateways and routers can be used to connect innumerable small networks to

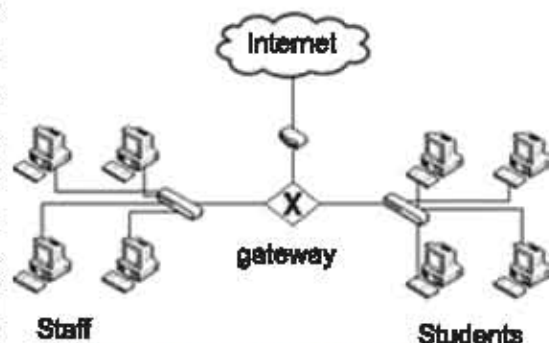


Figure 2.33: Usage of a Gateway

create a huge network. Gateways use Protocol Address Translation or PAT to identify networks. This is why it is also known as a protocol converter. It can filter data and can send data to a target IP address. It is slower than a router but uses better collision avoidance systems.

- **Network Interface Card:** In the recent past, computers needed a separate network interface card to connect it into a computer network. Now all the computers have the NIC built in with the system so it is almost obsolete nowadays.

2.5.5 Functions of Network

The main function of a computer network is communication and resource sharing. Functions of network devices inside a network is explained briefly in the following section:

1. Manage data communication between the network devices. Complete resource management.
2. Provide user access control and monitoring
3. Ensure fast and easy availability of information and resources

4. Create fast and effective communication with the world wide network.
5. Use in fields like education, healthcare, financial transactions, office work and many more.
6. Ensure the effectiveness and management of server machines
7. Provide back up data, ensure data safety and access.
8. Ensure the security and privacy of sensitive data.
9. Protect the system from unauthorized access.

2.5.6 Network Topology

Network topology refers to the manner in which the links and nodes of a network are arranged to relate to each other. Different network topologies grant different types of advantages to communication and resource sharing mechanisms. A network topology defines the physical connection pattern and how the data communication paths are arranged. A computer network can have devices other than computers. The connecting points of each computer or peripheral devices are called nodes in a network. The following topologies are generally used in computer networking systems;

1. Bus Topology
2. Ring topology
3. Star Topology
4. Tree Topology
5. Mesh topology
6. Hybrid topology

Bus Topology:

In this topology there is a central connection line where all the nodes or computer and other devices are connected. This central line is known as the Bus. Buses can be made of fiber optic cables and work as the backbone of a network. Buses have two terminators at each end.

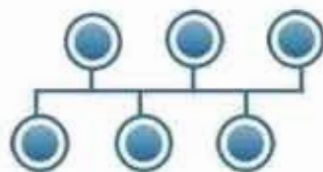


Figure 2.34: Bus topology

Each node of a network is individually connected to the bus. The data flow is bidirectional. When transmitting data, the sender device puts the data onto the bus with the receiver information. Every computer connected to the bus examines the data to determine who will receive the data. The receiver computer only receives the data and the others discard it.

Advantages of bus Topology

1. Bus topology is cost effective and the installation is simple.
2. Using repeaters or connectors one can easily increase the length of the backbone thus expanding the network.
3. New devices can be added to the network at any time.
4. Even if any node is disconnected from the network or is disabled, the network keeps running.
5. No centralised server is required.

Disadvantages of bus topology

1. Data transfer rate is slower.
2. If there is a problem with the bus the whole network suffers.
3. Adding computers to the network and expanding the network increases the network traffic and decreases network speed.
4. Chances of data collision.

Ring Topology

In ring topology the computer devices are connected in a circular fashion with one another to form a ring. In this network the first device and the last device are connected to form a ring. No central server is needed in this topology.

Each computer in this network gets equal opportunity to send and receive data. When a node sends a signal, it is transmitted to the node next to it. If the next node is the receiver it receives the data, if it is not the receiver it passes the signal to the next node until the receiver is reached.



Figure 2.35: Ring topology

Advantages of Ring Topology:

1. No centralised server is required.
2. Data flow is one directional so no chance of data collision.
3. Each computer gets equal priority in sending data.
4. Requires less cable so cost effective.

Disadvantages of Ring Topology

1. Data transfer rate is slower.
2. As the data flow is unidirectional and each computer receives and passes on the data until the destination is reached, if one computer device in the network is disabled. The whole network becomes disabled.
3. Adding or removing a computer device into the network disrupts the whole network.
4. Increasing the number of computer devices in the network increases the data transmission time too.
5. Complex softwares is needed to manage this topology.

Star Topology:

In star topology different computer devices and peripheral devices are connected to each other with a central hub or a switch. The devices in this topology use the hub or the switch to communicate and send/receive data between themselves. Data transmission requires less time and there is less chance of a data collision. The data flow is bidirectional. If there is a problem in any part of the network it is easily detectable through the hub or switch. Twisted pair cables are generally used to construct this topology.



Figure 2.36: Star topology

Advantages of Star Topology

1. Data transfer rate is faster
2. Less chances of data collision
3. Adding or removing nodes from the network does not disrupt the network.
4. The network keeps running even if one or more nodes are disabled or disconnected.
5. Increasing the number of network devices does not affect the data transfer rate.

Disadvantages of Star Topology

1. Problems with the hub or switch can disable the whole network
2. Each node requires cables of their own, so increases cost.
3. Two computers in a network can not directly communicate with each other.

Tree Topology

In Tree topology the network nodes are interconnected like branches of a tree. In this topology different levels of computer devices are connected with a central host computer or a server. Different levels of hierarchy are maintained while connecting different computer devices via hubs or switches. This is why it is also known as a hierarchical topology. In this system each computer acts as host computers for the next level of hierarchy. The last level is called the peripheral terminal as it does not have another level after it. A tree topology can be expanded. The data flow in this topology is bidirectional.



Figure 2.37: Tree topology

Advantages of Tree Topology

1. Can expand the network easily with expanding a branch or adding a new branch.
2. This provides a better system while constructing a large network.
3. The network does not face any disruption if one or more nodes are disconnected or disabled.
4. Data security is higher.
5. If a branch is disabled or disconnected the rest of the network can still be running.

Disadvantages of Tree Topology

1. If the root host server is disabled or disconnected the whole network is disabled.
2. This is more complex than other topologies.
3. This is more costly than other topologies.
4. If a computer in the network is disabled the devices connected to it in lower levels are disconnected too.

Mesh Topology

In mesh topology every device in the network is directly connected to every other device in the network. In this system every device can make faster and direct communication with other devices present in the network. This system does not need a centralised server computer. This connection between the computers is called peer to peer links. This topology is also known as a completely interconnected topology. This needs a lot of wiring and devices, so it is more costly.



Figure 2.38: Mesh topology

Due to its complex configuration it is not generally used in computer networking. If the network in mesh topology has number of nodes, in that case each node will need $n-1$ connections. So the number of wires needed is $n(n-1)/2$. Mesh topology is only used where the priority is data security and the dependability and reliability of a network. Defence sector and banking sectors use this kind of topology.

Advantages of Mesh Topology

1. Data transfer rate is faster.
2. Increasing computer devices in the network does not affect the data transmission speed.
3. The network keeps running even if one or more devices inside the network are disconnected or disabled.
4. Data transmission can continue even if one or more cables are disconnected or disabled.
5. No centralised server is required.

Disadvantages of Mesh Topology

1. This is more costly than any other topology
2. Installation and configuration of the network is complex.
3. Cost increases if a new device is added to the network.

Hybrid Topology

Hybrid topology accompanies star, ring, mesh and bus topologies. This is used when for some particular needs, one type of topology does not meet all of the requirements. The internet is based on hybrid topology. The Internet has all types of network topologies connected to it.

Advantages of Hybrid Topology

1. Network can be easily expanded using hubs or switches.
2. Troubleshooting of the network is easier.
3. If a part of the topology is disabled or disconnected, it does not affect other parts of the network.
4. As this is a combination of all other topologies it has the advantages of the other topologies too.

Disadvantages of Hybrid Topology

1. The maintenance cost is higher and the maintenance of this system is complex.
2. Installation and configuration of this system is complicated.
3. As this is a combination of all other topologies, it has the disadvantages of the other topologies too.

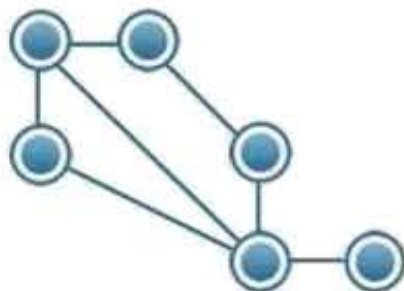


Figure 2.39: Hybrid topology

2.5.7 Cloud Computing

Simply put, cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use. It helps you lower your operating costs, runs your infrastructure more efficiently, and scales as your business needs change.

In the modern era cloud computing plays a very important role as most of the services available online are run on cloud. We use the internet to go into our social media and email accounts. We use these services for free. We can use these services because these companies have countless servers in many different countries around the globe catering to our needs. There are companies who provide you services against a payment. Both the free and premium services fall under the cloud computing systems. Cloud computing is not a technology, it is a business model in which the customers use their own computers or devices to access services, softwares, storage spaces from the providers who have servers situated anywhere in the ‘cloud’.

Cloud Computing can be categorized into three types:

Private Cloud: A singular organization has the ownership and control of the management of a cloud computing system here. This type of cloud is generally used and maintained by large organizations as though it is very expensive, it is more cost effective than managing multiple data centers across all of the organization's premises.

Public Cloud: The cloud services that are open to the public on the internet is known as the public cloud. Public cloud offers different services to the public which includes softwares, processing power, database, storage space and many more with or without payments. Companies like Amazon, Google, Microsoft have their own infrastructure and data centers around the world to make these types of public cloud services available to people all over the world.

Hybrid Cloud: As the name suggests, hybrid cloud is the combination of private and public or community clouds. Cloud services are combined into hybrid clouds to increase their efficiency and expand their coverage.

2.5.8 Advantages of Cloud Computing

The services provided by the cloud computing organizations can be divided into four types:

IaaS Infrastructure as a service: In this model service infrastructures are rented out via the cloud. Amazon's elastic computing cloud or EC2 is a popular model of this kind. Each EC2 server machine runs 1 to 8 virtual machines. These machines are rented out to the customers. Customers can install any kind of operating systems and softwares in these virtual machines and run application softwares of their choosing.

PaaS Platform as a Service: In this model rather than renting out virtual machines, computing platforms are rented out as services. This can include operating systems, development environment, execution environment, database, web servers and such. Customers can use this type of cloud services for application development without having to buy expensive computing systems on their own. Microsoft's Azure and Google's App Engine are prime examples of PaaS.

SaaS Software as a Service: In this model customers get access to the service application and their databases. So the customers do not have to use their own storage, processing power for the softwares or maintain and update it in anyways. Google Drive is a very popular SaaS system. Example of this model are Google App, Dropbox, Hubspot etc.

Network-Based Services (Naas: Network as a Service) : This is a model where customers rent network services from a cloud vendor instead of setting up their own network infrastructure. For example, the organization of Arjica and Partino provides services of WAN and VPN (Secure Virtual Private Network).

Cloud service users also get the following benefits:

Resource Flexibility and Scalability: Cloud services can cater to customers of any kind. Customers can increase or decrease their demand at any time and cloud services can cater to that instantly and remotely.

On Demand Services: Cloud services work as on demand service models. Customers can start or end their services at any given moment.

Pay as You Go: Customers do not have to pay in advance for any services. The payment will be according to the amount of the service taken. If the customer did not use a portion of a service he needs not pay for it.

Opportunity for Entrepreneurs: Cloud computing and cloud services have opened a vast amount of doors for the budding entrepreneurs. Now to start up a business you do not need your own hardware and most of the essential softwares and database systems are available for rent on the cloud. You do not need maintenance for cloud softwares, no need to manage or purchase license keys. Most of the cloud services and applications are easy to use and can be operated by people with moderate knowledge about the internet.

Cloud computing has really expanded the horizons for us in the field of information technology. But like all good things this system has some risks attached. Data ownership and privacy of information is a very sensitive issue in the cloud computing world. As the customers do not have a clue where his data is located or processed, it can create issues regarding data security and privacy.

Exercise

MCQ

1. What is the unit of data transfer?
 - a. Band Meter
 - b. Bandwidth
 - c. Data Transmission
 - d. Data Connection
2. A group SMS is -
 - a. Unicast
 - b. Multicast
 - c. Broadcast
 - d. Telecast
3. Which device can perform data filtering?
 - a. Hub
 - b. Switch
 - c. Repeater
 - d. Router
4. Bit synchronisation is -
 - i. A combined system to transmit bits
 - ii. The arrangement of bits of data and an extra bit.
 - iii. Extra bandwidth
 Which is true?
 - a) i and ii
 - b) i and iii
 - c) ii and iii
 - d) i, ii and iii
5. The purpose of a computer network is -
 - i. Sharing hardware resources
 - ii. Sharing software resources
 - iii. Connecting multiple computers
 Which one is true?
 - a) i and ii
 - b) i and iii
 - c) ii and iii
 - d) i, ii and iii

Read the following passage and answer the questions no 6 & 7:

One day Humaira went to his father's office and saw that his father gave a print command from his computer and another officer in that room gave a print command from his own computer. But the print out was made from the same printer. Humaira's father used his computer to talk with another officer out of the country.

6. The network described in the passage is -
 - i. LAN
 - ii. WAN
 - iii. MAN
 Which one is true?
 - a) i and ii
 - b) i and iii
 - c) ii and iii
 - d) i, ii and iii
7. In the described system one can have -
 - i. More services from a small device
 - ii. Easy communication with clients
 - iii. Limited work in a small area

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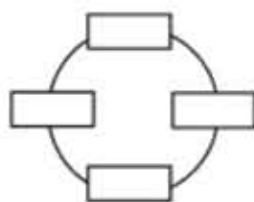


Figure 1



Figure 2

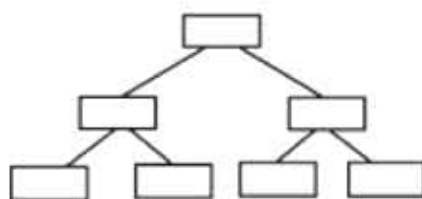


Figure 3

- a. What is Modulation?
 - b. Explain data transfer modes.
 - c. Explain the Network Topology In figure 1.
 - d. Between figure 2 and 3, what advantages and disadvantages will you get from applying each of these? Explain.
3. Shawpna was sharing data for free with his colleague Arif who sits on another floor. Meanwhile another colleague from the fifth floor asked for a file and Shawpna sent the file to his colleague without getting up from his seat. Shawpna also sent another file to one of his colleagues who is out of the country.
- a. What is full duplex?
 - b. Explain the transmission of data blocks as packets.
 - c. Explain the network Shawpna used to transfer files to Arif.
 - d. The second network is better than what was used with Arif, explain.
4. An educational institution brought computers from different floors under one network. But after a few days one computer machine was not running and that disrupted the whole network.
- a. What is a Bridge?
 - b. Explain the concepts of wireless communication.
 - c. What Network Topology did the educational Institution use?
 - d. To avoid disruption, what Network Topology should they use? Explain your answer.